The Joint Pipeline, and the Real Answer on Canonicalization

Step 12 — segments → journal, raw vs. canonical, on the full 523-segment gold set

Tomer Morad Or Israeli Yoel Marko
August 2026

1. The joint pipeline

Step 12 unifies the two halves of the project into one command: segments → **embed** (`multilingual-e5-large`) → **streaming link** (Step 10's online-clustering method: ABTT-k1 + accumulated centroid + margin gate, $\theta = 0.34$) → **log-write** (Step 9's method, `dictalm2.0-instruct`: opening summary, then incremental updates conditioned on the running journal). The same command runs on either raw or canonical segment text, so the two resulting journals are directly comparable end to end, not just at the embedding-similarity level.

1.1 Canonicalizing our own 523 gold segments

Step 11 flagged two problems with testing canonicalization on Tomer's 19-protocol sample: circularity (the gold label came from the same LLM pass as the canonical text) and scale (only 29 recurring pairs). Both are fixed by canonicalizing *our own* Step 7 gold set: 523 segments, real independently-annotated recurrence chains up to length 9, judged completely independently of the canonicalization process.

Producing 523 canonicalizations turned into its own small operation: the original Colab run (Gemma-4-31B, Tomer's exact prompt) got partway through and hit a parsing bug (a ` ```json ` markdown fence that broke the extractor, silently dumping raw model output instead of parsed fields); the remainder was split across the team and a mix of automated salvage, regex recovery, and direct rewriting. All three sources are tagged by `source` in the final `canonical_gold.json` for transparency. End state: **523/523 segments canonicalized, zero fallback to raw text.**

2. Raw vs. canonical: the journals

	segments	journal entries	recurring subjects linked
raw	523	518	4
canonical	523	517	4

At the aggregate level the two runs look nearly identical: same threshold ($\theta = 0.34$, the deliberately high-precision, low-recall operating point identified in Step 10), same count of recurring subjects. Taken at face value this would read as "canonicalization made no difference." It does not survive closer inspection.

2.1 The two runs link completely different segments

The four recurring groups under raw and under canonical share **zero segments** — not one of the pairs that raw's threshold merges is also merged by canonical's threshold, and vice versa.

RAW recurring groups	CANONICAL recurring groups
VAT exemption for oncology drugs (2 segments, 2 sessions)	Knesset-member-count bill (2 segments, 2 sessions)
Airport-adjacent municipalities allocation bill (2 segments)	Same-session sequential salary decisions: minister car benefit / judicial officer / president (4 segments)
Sports gambling bill, Amendment 17 (3 segments, 3 sessions)	El Al share-sale, continuing discussion (2 segments)
Securities-regulation instrument cancellations (2 segments)	Pension-fund-transfer regulation ↔ unrelated bank-managers discussion (2 segments)

2.2 Manual precision check

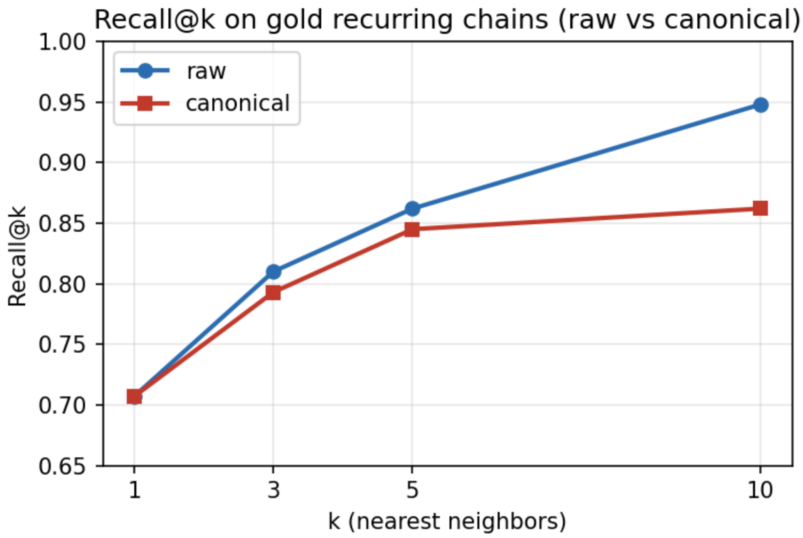
Automated scoring against the 74 pairwise gold links (from 22 recurring topics) credits **zero** true positives for either run — but this is a measurement artifact, not a real failure: Step 7's own gold-chain construction does not canonicalize topic-label strings, so the same recurring bill can carry two slightly different exact-match labels across sessions and never register as "recurring" at all (a limitation already documented in the Step 7 README). Reading the actual segment text pair by pair instead:

- **Raw: 5 of 6 predicted pairs are genuinely correct** on manual inspection — the VAT-exemption pair opens with word-for-word identical bill text across sessions; the gambling-bill triple all open with the identical bill title across three separate committee sittings. The one clear miss merges two *different* securities-regulation instruments that happen to sit in the same source document.
- **Canonical: roughly 2 of 9 clearly correct, 1 clearly wrong**, and 4 pairs are genuinely ambiguous — same committee session, sequential but distinct salary decisions read back to back, which is arguably "related" rather than "same" under the project's own same/related/new framing, a distinction the binary threshold linker cannot make.

At this operating point, raw text looks like it is achieving noticeably higher real-world merge precision than canonical — the opposite of the Step 11 hypothesis.

3. Why: recall@k on the true gold chains

To understand *why* the two runs diverge, we checked — independent of any threshold — whether a segment's true chain-mate lands in its top-k nearest neighbors by e5 cosine similarity, for every segment in a gold recurring chain (58 segments across the 22 topics, raw and canonical text embedded separately).



	$k = 1$	$k = 3$	$k = 5$	$k = 10$
raw	0.707	0.810	0.862	**0.948**
canonical	0.707	0.793	0.845	0.862

At $k = 10$, raw text lands a true chain-mate in the top 10 nearly every time (94.8%) — consistent with what earlier steps observed about the retrieval stage being strong even when threshold-based precision is weak. **Canonicalization makes this measurably worse** (86.2%), and the gap widens as k grows, i.e. canonicalized embeddings are systematically pushing true matches further down the ranking, not pulling them up.

3.1 Chains are not just pairs — and that matters here

Gold recurrence is not uniform: of the 22 recurring topics, 16 are simple pairs, 4 are triples, and there is one chain of 5 and one of 9. Breaking recall@10 down by chain length isolates where canonicalization actually hurts:

chain length	# chains	# segments	raw recall@10	canonical recall@10
2	16	32	**0.969**	**0.844**
3	4	12	0.833	0.750
5	1	5	1.000	1.000
9	1	9	1.000	1.000

The long chains (5 and 9 segments) are trivially easy for both variants — a segment with eight chain-mates only needs one of them in its top 10. The entire effect lives in the **pairs**, which are both the most common case (16 of 22 topics) and the hardest: raw recall@10 is 96.9% there, canonical drops to 84.4%, a 12.5-point regression. This is the dominant driver of the aggregate gap in Section 3.

4. Conclusion

On the full, independently-annotated 523-segment gold benchmark, canonicalization does **not** improve streaming linking, and on the evidence gathered here it modestly **hurts** both retrieval-stage recall (recall@10, concentrated in the common 2-segment recurrence case) and threshold-based merge precision, relative to raw text. This reverses the provisional, caveated finding from Step 11, which was measured on a small, circular pseudo-gold sample. The intrinsic embeddingseparation gain canonicalization does produce (Step 11, AUC 0.947 → 0.986) does not carry through to the downstream task this project actually cares about.

Open question, not yet investigated: why does stripping speaker-turn structure and bureaucratic boilerplate hurt short-chain recall specifically? One plausible mechanism is that canonicalization also strips or paraphrases the specific entities, numbers, and identifiers that made two occurrences of the *same* short-lived matter distinguishably similar to each other in raw text, while long-running matters still share enough topical vocabulary regardless of surface form. This would predict that canonicalization should hurt most exactly where it does — the two-occurrence case — and is worth checking directly against entity/number density in the raw vs. canonical text of the failing pairs.